

Recall

Linear Combination of vectors

$V \longrightarrow$ vector space

$$S = \{v_1, v_2, \dots, v_k\} \subseteq V$$

Linear span/
generator set

Linear Combination of vectors in S

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k \quad ; \quad \alpha_i \in F$$

Linearly dependent set

A subset S is LD if $\exists$ non-zero scalars $\alpha_1, \alpha_2, \dots, \alpha_k$

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = 0$$

Linearly independent set

A subset S in $L.I$ if

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = 0$$

$$\Rightarrow \alpha_1 = \alpha_2 = \dots = \alpha_k = 0$$

Question $\mathbb{R}^2$

$\{(0,0)\}$ is L.I. set??

$$1 \cdot (0,0) = (0,0)$$

$\{(0,0), (1,0), (0,1)\}$ is linearly dependent set

$$1 \cdot (0,0) + 0 \cdot (1,0) + 0 \cdot (0,1) = (0,0)$$

* Let V be a vector space

$\{\alpha_1 v_1, \alpha_2 v_2, \dots, \alpha_k v_k\}$ is a linearly dependent set.

$\mathbb{R}^2$

$\{(0,0)\}$

generator $\longrightarrow (0,0)$

$\{(1,0)\}$

linear span $\longrightarrow x$ axis ✓

$\{(2,5)\}$

linear span $\longrightarrow$ straight line passing $(0,0)$ and $(2,5)$ ✓

$\{(1,0), (2,5)\}$

linear span

$\mathbb{R}^2$

= full vector space

Minimal spanning set

Minimal spanning set

$$\{(1,0), (2,5)\} \xrightarrow{\text{span}} (\mathbb{R}^2) = \text{full vector space}$$

$$(x,y) = \left(x - \frac{2y}{5}\right)(1,0) + \frac{y}{5}(2,5)$$

Minimal spanning set : Let V be a vector space over F .

A subset S is called minimal spanning set of V if linear span of S is V and no proper subset of S spans full vector space V .

Minimal spanning set = Maximal linearly independent set.

Basis of vector space

$$\{(1,0), (2,5)\}$$

$$\mathbb{R}^2$$

$$\{(1,0), (2,5), (x,y)\}$$

$$\left(x - \frac{2y}{5}\right)(1,0) + \frac{y}{5}(2,5) - 1(x,y) = (0,0)$$

Basis :

Let V be a vector space over F . A subset B is called Basis of V if

(i) Linear span of B is V .

(ii) B is linearly independent set.

$$\mathbb{R}^2$$

$$\{(1,0), (2,5)\}$$

is Basis of $\mathbb{R}^2$

$$(x,y) = \left(x - \frac{2y}{5}\right)(1,0) + \frac{y}{5}(2,5) \quad (\text{spanning set})$$

L.I. set

$$\begin{cases} \alpha_1(1,0) + \alpha_2(2,5) = (0,0) \\ (\alpha_1 + 2\alpha_2, 5\alpha_2) = (0,0) \\ \alpha_1 + 2\alpha_2 = 0, \quad 5\alpha_2 = 0 \end{cases} \rightarrow \underline{\alpha_1 = 0, \alpha_2 = 0}$$

- $\{(1,0), (0,1)\}$ $(x,y) = x(1,0) + y(0,1) \rightarrow$ Linear span
- $\{(2,0), (0,2)\}$ $\left. \begin{aligned} \alpha_1(1,0) + \alpha_2(0,1) &= (0,0) \\ \alpha_1 &= 0, \alpha_2 &= 0 \end{aligned} \right\} \rightarrow$ Linearly independent.
- $\{(1,0), (1,1)\}$

* Basis of a vector space is NOT unique

$$\begin{aligned} \mathbb{R}^2 & \quad \{(1,0), (0,1)\} \\ \mathbb{R}^3 & \quad \{(1,0,0), (0,1,0), (0,0,1)\} \\ \mathbb{R}^n & \quad \left\{ \overset{e_1}{(1,0,\dots,0)}, \overset{e_2}{(0,1,0,\dots,0)}, \overset{e_3}{(0,0,1,0,\dots,0)}, \dots, \overset{e_n}{(0,\dots,0,1)} \right\} \end{aligned}$$

$\mathbb{R}^n \ni e_i = \begin{cases} 1 & i\text{th place} \\ 0 & j\text{th place} \neq i, 1 \leq j \leq n \end{cases}$

} Basis of $\mathbb{R}^n$

$M_2(\mathbb{R})$ is v.s $\mathbb{R}$

$$B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

B spans $M_2(\mathbb{R})$.

$$\alpha \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \beta \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + \gamma \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + \delta \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

B is LI set

$$\begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \underline{\alpha=0, \beta=0, \gamma=0, \delta=0}$$

• $\{0, 1, x, x^2, \dots, x^n\}$ polynomial of degree $\leq n$

$P_n(\mathbb{R}) = \left\{ \text{set of polynomial of degree } \leq n \right.$
 $\left. \text{with } n \right\}$

$$= \left\{ \underline{a_0} + \underline{a_1}x + \dots + a_n x^n \mid a_i \in \mathbb{R} \right\}$$

Basis = $\{1, x, x^2, \dots, x^n\}$ Exercise

$\mathbb{R}^2$

$\{(1,0), (2,5)\}$

$\{(1,0), (0,1)\}$

$\{(2,0), (0,2)\}$

$\{(1,0), (1,1)\}$

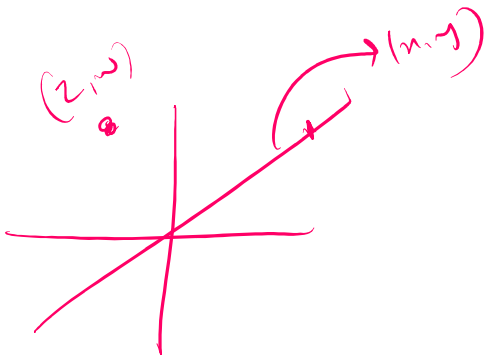
→ Each of them
contains 2
L.I. vectors.

$\{(0,0)\} \rightarrow \{(0,0)\}$

$\{(x,y) \mid (x,y) \neq (0,0)\}$

↓ gen

line passing through (x,y) & $(0,0)$



$\{(x,y), (2,w)\} \rightarrow$ L.I.

$\mathbb{R}^2$

$\{(x,y), (2,w), v_3\} \rightarrow$ L.D.

Dimension

Let V be a vector space over F .
 Let B_1 and B_2 be two bases of vector
 space then $|B_1| = |B_2|$ $\rightarrow$ Cardinality

space then $|B_1| = |B_2|$ ← Cardinality of B_1 ← Cardinality of B_2

• Number of elements/vectors in the basis of a vector space is called Dimension of vector space

• If $\dim(V) = n$, then any subset of vector space V containing m elements, where $m > n$, is a L.D. set.

• $\dim(\mathbb{R}^2) = 2$ $\dim(\mathbb{R}^3) = 3$

• $\dim(\mathbb{R}^n) = n$ $\dim(M_2(\mathbb{R})) = 4$

$\dim(P_n(\mathbb{R})) = n+1$

$\dim(M_{n \times m}(\mathbb{R})) = nm$

$\dim(M_n(\mathbb{R})) = n^2$

• How to find Basis of a vector space?

→ L.I. set which spans V

① start with non-zero vector $\{v_1\} \rightarrow$ L.I. set.

② $L.S\{v_1\} = V$ (stop here $\{v_1\}$ is basis of V)

→ If $L.S\{v_1\} \neq V$
 $\dots \subset V \setminus L.S\{v_1\}$

$\rightarrow$ If $L.S\{v_1\} \neq V$
 then $\exists v_2 \in V \setminus L.S\{v_1\}$
 $\{v_1, v_2\}$ is L.I. set.

③ $L.S\{v_1, v_2\} = V$??

Yes $\swarrow$
 $\{v_1, v_2\}$ is basis
 No $\searrow$
 $v_3 \in V \setminus L.S\{v_1, v_2\}$
 $\{v_1, v_2, v_3\}$ is L.I. set

④ $L.S\{v_1, v_2, v_3\} = V$

Example 1 - Find the Basis of $\mathbb{R}^4$
 containing $\{ \overset{v_1}{(1, 1, 0, 0)} \} = S_1$
 $L.S(S_1) = \{ \underline{(x, x, 0, 0)} \mid x \in \mathbb{R} \} \neq \mathbb{R}^4$

$v_2 \in \mathbb{R}^4 \setminus L.S(S_1)$

$v_2 = (0, 1, 0, 0)$

$S_2 = \{ (1, 1, 0, 0), (0, 1, 0, 0) \}$ $\rightarrow$ $L.S_2$

$\beta(1, 1, 0, 0) + r(0, 1, 0, 0) = (0, 0, 0, 0)$
 $\beta = 0$
 $\beta + r = 0 \rightarrow r = 0$

$\rightarrow (x, x, 0, 0) \in \mathbb{R}^4$

$$\beta + \gamma = 0$$

$$L.S(s_2) = \left\{ \alpha \underset{v_1}{(1, 1, 0, 0)} + \beta \underset{v_2}{(0, 1, 0, 0)} = \underset{(a, b, c, d)}{(\alpha, \alpha + \beta, 0, 0)} \mid \alpha, \beta \in \mathbb{R} \right\} \neq \mathbb{R}^4$$

$$v_3 \in \mathbb{R}^4 \setminus L.S(s_2)$$

$$v_3 = (0, 0, 1, 0)$$

$$S_3 = \{ (1, 1, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0) \}$$

check S_3 is L.I. set.

$$L.S(S_3) = \left\{ (\alpha, \alpha + \beta, \gamma, 0) \mid \alpha, \beta, \gamma \in \mathbb{R} \right\} \neq \mathbb{R}^4$$

$$v_4 \in \mathbb{R}^4 \setminus L.S(S_3)$$

$$v_4 = (0, 0, 0, 1)$$

$$S_4 = \{ (1, 1, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1) \}$$

↘ L.I. set.

$$L.S(S_4) = \mathbb{R}^4$$

$$(x, y, z, w) = x(1, 1, 0, 0) + (y - x)(0, 1, 0, 0) + z(0, 0, 1, 0) + w(0, 0, 0, 1)$$

$$\mathbb{R}^2$$

$$S = \{ (1, 0), (1, 1), (2, 5), \cancel{(0, 0)} \} \text{ is spanning set for } \mathbb{R}^2$$

$S = \{ (1,0), (1,1), (2,5), \cancel{(1,1)} \}$ let $\mathbb{R}^2$

Find basis B of $\mathbb{R}^2$, which B.C.S.

$\{ \underline{(1,0)}, \underline{(1,1)}, \cancel{(2,5)} \} \rightarrow$ linearly
 depen

$$(2,5) = (-3)(1,0) + 5(1,1)$$

$\{ (1,0), (1,1) \} \xrightarrow{\text{L.E.}} \text{spanning } B$